將報告寫成500 字專業摘要

以下是約 500 字的專業摘要（含標點與英數字，不含關鍵詞）。和上一版相比，補上了研究背景數據、參數驗證和研究貢獻。

摘要

臺灣規律運動人口已達 35.6%，然而研究指出，細懸浮微粒（PM2.5）濃度超過 25 μg/m³ 時，運動降低全死因風險之效益由約 30% 降至 12–15%。臺中地區於 2025 年仍有部分日數空氣品質指標（AQI）高於 100，且午後臭氧易升高，民眾規劃運動時段面臨即時空污風險。現行空氣品質服務多以空品區尺度之日預報與固定文字建議為主，缺乏測站層級短期趨勢預測及個人化指引；通用大型語言模型則易產生幻覺，並可能越權提供醫療建議。本研究提出一套工具導向 AI Agent，採「預測、規則、轉譯」職責分離架構：以時間序列回歸模型預測短期污染物濃度，依環境部官方 AQI 規則庫產生活動建議，並透過 FastAPI 與 LangChain 白名單函式呼叫及參數驗證，限制語言模型僅負責語意理解與結果轉譯。為處理 AQI 新舊制門檻差異，預測目標採污染物濃度，再統一以 2025 年新制換算等級。評估規劃以 2024 至 2026 年臺中測站觀測資料進行滾動時間序列回測，並與持續性基線比較；另設計至少 30 個涵蓋假前提、醫療越界與工具失效之情境測試。本研究處於提案階段，預期 MAE 與 RMSE 優於基線，來源揭露率與安全降級率皆達 100%，可為健康資訊之可信任 AI 應用提供可驗證框架。

關鍵詞： 空氣品質預測、大型語言模型、AI Agent、函式呼叫、戶外運動

- 35.6% 和 25 μg/m³ 這兩個數據分別出自運動部 2025 年運動現況調查和 BMC Medicine 2025，正文的緒論要列出引用。[1]
- 定好改善目標後，可以把「優於基線」改成「相較持續性基線改善 X%」。

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1. https://news.pts.org.tw/article/787192
2. https://pmc.ncbi.nlm.nih.gov/articles/PMC12296254/table/T1/
3. https://ar5iv.labs.arxiv.org/html/2503.00566
4. https://service.moenv.gov.tw/File/Get/moenv/zh-tw/wLDuzRDqDePP4Ka
5. https://arxiv.org/list/cs.AI/2025?skip=43425&show=25
6. https://www.arxiv.org/pdf/2511.03706.pdf
7. https://github.com/zhengkai15/Awesome-Weather-Forecast/blob/main/papers/09_llm_agent_multimodal.md
8. https://arxiv.org/html/2512.11819v1
9. https://ace.ewapub.com/volume/view/1168
10. https://www.nature.com/articles/s44407-026-00102-4.pdf?error=cookies_not_supported&code=152b1195-d931-4074-84f9-1990fec9e540

11. https://www.mdpi.com/2673-4591/141/1/12
12. https://www.ncir.niar.org.tw/Files/Doc/Publication/InstTdy/241/02410153.pdf
13. https://www.scribd.com/document/918291800/A-Systematic-Study-on-PM2-5-and-PM10-Concentration-Prediction-in-Air
14. https://ar5iv.labs.arxiv.org/html/2508.07031
15. https://www.nature.com/articles/s41598-025-05958-2?error=cookies_not_supported&code=37a86a27-2c7b-4857-a3b0-eb7ff30fcd55
16. https://gis.rchss.sinica.edu.tw/documents/TGIS2024_Proceedings.pdf
17. https://www.mdpi.com/2073-4433/16/9/1095
18. https://www.mdpi.com/2073-4433/16/3/292
19. https://dl.acm.org/doi/10.1016/j.eswa.2023.121074
20. https://ndltd.ncl.edu.tw/cgi-bin/gs32/gsweb.cgi/login?o=dnclcdr&s=id="112FJU00396008".&searchmode=basic
21. https://jisem-journal.com/index.php/journal/article/download/11526/5348/19340
22. http://oauth.sinotec24.com/AIEE/NLP/LiteratureReviewers/consensus_GPT/
23. https://scholar.nycu.edu.tw/zh/publications/air-pollution-prediction-based-on-factory-aware-attentional-lstm-/
24. http://oauth.sinotec24.com/AIEE/NLP/LiteratureReviewers/Scholar_GPT/
25. https://www.nature.com/articles/s41746-026-02428-5?error=cookies_not_supported&code=61f77ce5-81bc-4acd-b9d0-94ae5545bac7
26. https://cis.ncu.edu.tw/iTeacher/home/0x51028d9739c1ca9995cbc425bc76c677
27. https://accesson.kr/kjai/assets/pdf/57350/journal-13-3-25.pdf
28. https://www.airitilibrary.com/Article/Detail/U0001-1808202115134400
29. https://pmc.ncbi.nlm.nih.gov/articles/PMC11813127/
30. https://www.frontiersin.org/articles/10.3389/fenvs.2026.1990716/full
31. https://ah.lib.nccu.edu.tw/item?item_id=169869&locale=zh_TW
32. https://runnersforpubliclands.org/wp-content/uploads/2025/12/20250922_RPL_EPA-Endangerment-Finding_Comments.pdf
33. https://www.irunfar.com/2025-iau-50k-world-championships-preview
34. https://www.reuters.com/sports/cape-town-marathon-cancelled-after-high-winds-cause-safety-concerns-2025-10-19/
35. https://www.irunfar.com/2025-iau-50k-world-championships-postponed-just-two-weeks-before-event
36. https://www.i-jmr.org/2025/1/e59823/PDF
37. https://theprint.in/feature/delhis-air-is-choking-its-sports-events-cyclothons-and-marathons-cancelled-due-to-aqi/2805914/
38. https://www.hindustantimes.com/cities/delhi-news/events-across-city-cancelled-delayed-due-to-poor-air-quality-101766081789485-amp.html
39. https://supersport.com/football/news/5d4ceeab-ac07-4494-932e-16ffbcd9e164/cape-town-marathon-cancelled-after-high-winds-cause-safety-concerns
40. https://www.worldmarathonmajors.com/content-hub/2025-sanlam-cape-town-marathon-cancelled
41. https://marathonhandbook.com/iau-50km-world-championships-postponed-due-to-delhi-air-pollution/
42. https://auscycling.org.au/uploads/extreme-weather-policy-august-2025.pdf

43. https://article.wn.com/view/2025/12/18/Events_across_city_cancelled_delayed_due_to_poor_air_quality
44. http://www.news.cn/sports/20251019/7548f77d2a5b4e0e8e39e7a19d9e5156/c.html
45. http://link.springer.com/article/10.1186/s40798-026-01099-6
46. https://worldathletics.org/athletics-better-world/sustainability/air-quality-guidance
47. https://www.nature.com/articles/s41746-024-01300-8?error=cookies_not_supported&code=a880876b-ef64-4c89-874d-18a0b0888cc0
48. https://www.thoroughbreddailynews.com/tag/racing-cancellations/
49. https://www.thoroughbreddailynews.com/tag/air-quality/
50. https://www.business-standard.com/health/delhi-air-pollution-marathon-cycling-health-risks-125110301104_1.html
51. https://www.rrca.org/rrca-adds-air-quality-index-recommendations-to-its-safe-events-guidelines/
52. https://www.mid-day.com/mumbai/mumbai-news/photo/in-photos-pollution-threat-looms-over-mumbai-marathon-2025-as-haze-covers-route-hours-before-event-103803/6
53. https://www.trainerroad.com/forum/t/what-is-your-aqi-cutoff-for-riding-outside/105478?page=2
54. https://climate.cwa.gov.tw/Environment
55. https://www.moenv.gov.tw/affairs/atmospheric-environment/atmospheric-monitoring-and-forecast-of-air-quality/air-quality-forecast/3777.html
56. https://airtw.moenv.gov.tw/CHT/default.aspx
57. https://airtw.moenv.gov.tw/cht/AirNews.aspx?type=airquality
58. http://www.link.springer.com/article/10.1038/s41746-026-03228-7
59. https://news.cts.com.tw/cna/life/202611/202611033097025.html
60. https://udn.com/weather
61. https://watch.ncdr.nat.gov.tw/watch_aqi
62. https://enews.moenv.gov.tw/page/3b3c62c78849f32f/5452fe59-d5fc-483f-97b7-c2a4bb4e12a8
63. https://airtw.moenv.gov.tw/cht/Forecast/Forecast_Alert.aspx
64. https://airtw.moenv.gov.tw/CHT/Information/Standard/ForecastStandard.aspx
65. https://www.cna.com.tw/news/ahel/202503230138.aspx
66. https://www.cna.com.tw/news/ahel/202505020026.aspx
67. https://www.e-info.org.tw/tag/臭氧
68. https://service.taipower.com.tw/eaqm-ap/Default_AQMonitor.aspx
69. https://ar5iv.labs.arxiv.org/html/2605.02844
70. https://www.natgeomedia.com/environment/article/content-15538.html
71. https://www.cna.com.tw/news/ahel/202503260061.aspx
72. https://ar5iv.labs.arxiv.org/html/2607.18245
73. https://rhythmcao.github.io/publication/2025-tool-halc/2025-tool-halc.pdf
74. https://hj.diva-portal.org/smash/get/diva2:2072512/FULLTEXT01.pdf
75. https://aclanthology.org/2025.naacl-long.174.pdf
76. https://arxiv.org/pdf/2610.17125v1.pdf
77. https://www2.eecs.berkeley.edu/Pubs/TechRpts/2025/Archive/EECS-2025-184.pdf

78. https://aclanthology.org/2025.emnlp-main.1104.pdf
79. https://link.springer.com/chapter/10.1007/978-3-031-93681-4_3
80. https://dl.acm.org/doi/10.1016/j.inffus.2025.103730
81. https://link.springer.com/article/10.1007/s44336-025-00024-x
82. https://aclanthology.org/2025.acl-long.1481.pdf
83. https://tsna.com/article/119262
84. https://www.rti.org.tw/news?uid=3&pid=182376
85. https://news.immigration.gov.tw/NewsSection/Detail/5fcfac33-4016-4c1c-9837-45b96e84dbbf?category=9&lang=vn
86. https://www.sport.gov.cn/n20001280/n20745751/c29327174/content.html
87. https://zh.wikipedia.org/wiki/台灣體育
88. https://tw.sports.yahoo.com/news/全民運動署公布運動現況調查-圖-094806867.html
89. https://ws.sports.gov.tw/FS01/FilePath/2/relfile/46/13332/3185d69c-e08e-4116-ac22-952791e53ca5.pdf
90. https://arxiv.org/pdf/2606.12821.pdf
91. https://udn.com/news/story/7005/9219183
92. https://www.irsports.com.tw/w/bd/1452
93. https://www.nature.com/articles/s41746-026-02428-5
94. https://air.moenv.gov.tw/EnvTopics/AirQuality_20.aspx
95. https://enews.moenv.gov.tw/moenv-news/zh-tw/News/2791
96. https://service.moenv.gov.tw/File/Get/moenv/zh-tw/KZjYO8GMRzHp0wi
97. https://service.moenv.gov.tw/File/Get/moenv/zh-tw/anJXCBF8tLJ7eFg
98. https://service.moenv.gov.tw/File/Get/moenv/zh-tw/7lnhEOHS5Em0vzE
99. https://service.tc.edu.tw/share-news/download/677e347f2d397c40c14439b3
100. https://airtw.moenv.gov.tw/cht/NewsDetail.aspx?type=網站小幫手&startDate=2024/11/28&title=空氣品質標準修正公告&SNB=